

## David M. Noll

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### Technology Skills

**Front End:** TypeScript, React, Vue, Redux, Prisma, Jest, Cypress

**Back End:** Python, Node.js, Django, Flask, PHP; Some Scala, Haskell, Java, Rust

**AI:** LLM Integrations, Clustering with Embeddings, prototyped semantic search

**DevOps:** AWS, Azure, Docker, SQL, Linux; Some Terraform, Kubernetes, AWS CDK, Ansible

### Experience

#### DevOps Engineer, Stottler Henke Associates Inc.

2025

##### AI Research & Prototyping

- Contributed to funded feasibility research proposal on recursive decomposition with LLM's and static analysis for automated refactoring and containerization.
- Conducted Phase 1 SBIR research exploring modularization and containerization of codebases
- Prototyped local log analysis tool for LLM analysis of logs using local Ollama instance to detect anomalies.

##### AI Infrastructure & Platform

- Research on-premises LLM hardware requirements, created proof of concept of internally hosted LLM for on-prem coding assistant & investigate limitations
- Create & Maintain API access to AWS Bedrock.
- Develop & Maintain sandbox container to provide LLM coding assistant access

##### Compliance, DevOps and infrastructure

- Automated tooling including AWS billing reports, Backstage portal for services with integration with on premises Gitlab installation, and health checks.
- Develop infrastructure as code for On-prem and AWS using CDK and Ansible.
- Research and development & prototyping of compliant on premises Kubernetes installation.
- Research, purchase, and test deployment of package firewall service.
- Manage & assist in compliance audits of internally hosted developer tooling (e.g. GitLab)

#### Developer, Fosforescent

2023-2024

##### Solo bootstrapped project for a collaborative organization & workflow system

- Designed a graph rewriting system for collaborative human + agent workflow management, combining functional programming concepts with practical task management.
- Implemented system using OpenAI API to recursively generate graph-based workflows.
- Prototyped semantic search with Pinecone and OpenAI embeddings API.
- Created TypeScript & React UI library to allow navigating & drag-and-drop editing of task graph.
- Created multiple iterations of backend including Rust web server running Stripe integration, and Node.js version with content-addressed node storage
- Developed CI/CD pipeline with GitHub Actions, infrastructure as code repository using Terraform, automated Kubernetes deployments using ArgoCD, with cloud databases
- Researched graph rewriting systems, combinator systems, process calculi, and type systems.

#### Lead Full Stack Engineer, Dojo.co

2021-2023

##### Led 3 other FT devs on ~10 person team doing analysis and visualization of workplace interaction data

- Designed and implemented a CI/CD pipeline in Heroku & GitHubActions and added an entire test suite using Cypress, Pytest, and Jest tests.
- Introduced Docker & Docker Compose local development & testing environment.

- Created & maintained production, testing and dev environments in Heroku, AWS, passing rigorous security review processes.

#### *Features*

- Improved performance and maintainability through multiple phases of refactoring front end and back end, including adding progressive loading to d3 force graph visualization, improving DB schema.
- Increased performance of endpoint from several minutes to under 10 seconds for large projects, increasing feasible project size by 100x+.
- Added AWS Lambda function with AWS Serverless Application Module to handle log drain.
- Designed & implemented concept of groups, group member, node hierarchy.
- Implemented new admin UI to conform to JWT auth requirement of 3rd party pen test.

#### *Management*

- Led remote team of 1 FT engineer as well as 2 regular full time contractors.
- Responsible for validating feature implementation plans, code review, quarterly feedback.

### **Senior Full Stack Engineer, *Care.Coach***

2018-2021

*First full time hire at Telehealth assistant based around remote caregiver-powered animated avatar*

- Designed & created Django-based internal service to house updated schema + machine learning models & integrations as well as collected, monitored & analyzed performance metrics and logs for reporting.
- Upgraded video streaming architecture from Flash/RTMP to HTML5/WebRTC by implementing web client, server configuration, guiding implementation of Android mobile client.
- Upgraded security to meet standards of major client 3rd party security audit.
- Integration with hospital EMR & our cloud-hosted SQL Server DB using Java Spring.

#### *DevOps*

- Created & maintained production, testing and dev environments in Azure.
- Completed migration of production & staging infrastructure from dedicated Linux server to Azure with minimal downtime.

### **Independent Contractor, *David M Noll LLC*, Louisville, KY**

2014-2015, 2017-2018

- Created web applications in PHP, JavaScript, AWS EC2, and shared hosting.
- Designed and implemented Python desktop application for cryptocurrency trading.

### **Interactive Developer, *JUMP by LEAP agency*, Louisville, KY**

2015-2016

- Development with WordPress, Laravel, or PHP/HTML, MySQL, JavaScript.

### **IT Consultant, *Derby City IT*, Louisville, KY**

2012-2013

### ***Projects***

**Robo1:** web app to allow group tele-operation of robots and simulation environments.

**GraphLang:** various experiments with graph rewriting programming language concepts.

**Netcode:** A neuron-wise neural net library in python for exploring backpropagation and node-level visualization.

**ChkFlow:** a React/typescript component editor/view for tree data inspired by Workflowy's interface.

**Pixel Time:** a dApp using typescript & solidity, web3.js that allows users to collaboratively edit an image.

**GLTrader:** desktop app using python for real-time monitoring of cryptocurrency market data via Bittrex API.

### ***Education***

**Rensselaer Polytechnic Institute**, M.S., Global Enterprise Management

2010

**Case Western Reserve University**, B.A., Cognitive Science

2008